H/w N1 Kurbatov Daniil 246

N1

uns: 0: 000000 = 0

13: 001101 = $2^3+2^2+2^0=13$

24: 011000 = $2^4+2^3=24$

63: 111111 = $2^5+2^4+2^3+2^2+2^1+2^0=63$

Signed:

16: 010000 = $2^4=16$

-2: since 2 = 000010 → invert → 111101 → add 1 → 111110

31: 011111 = $2^4+2^3+2^2+2^1+2^0$

-32: since 32 = 100000 → invert → 011111 → add 1 → 100000

N2

- 000101 = $2^2+2^0=5$ both signed and unsigned
- 101011 = $2^5+2^3+2^1+2^0=43$ unsigned and $-32+2^3+2^1+2^0=-21$ signed
- 111111 = $2^5+2^4+2^3+2^2+2^1+2^0=63$ usigned and $-32+2^4+2^3+2^2+2^1+2^0=-1$ signed
- 100000 = $2^5=32$ unsigned and -32 signed

N3

$7_{10} \to 07_{16}$

$240_{10} \to F0_{16}$

$171_{10} \to AB_{16}$

$126_{10} \to 7E_{16}$

N4

$0x3C_{16} = \underbrace{0011}_{3}\underbrace{1100}_{C}{}_2$

$0x7E_{16} = \underbrace{0111}_{7}\underbrace{1110}_{E}{}_2$

$0xFF_{16} = \underbrace{1111}_{F}\underbrace{1111}_{F}{}_2$

$0xA5_{16} = \underbrace{1010}_{A}\underbrace{0101}_{5}{}_2$

N5

1) 0011.1100 → invert → 11000011 → add 1 → 1100.0100 (0xC4)

2) 0111.1110 → invert → 1000 0001 → add 1 → 10000010 (0x82)

3) 1111.1111 → invert → 0000.0000 → add 1 → 00000001 (0x01)

4) 1010.0101 → invert → 0101 1010 → add 1 → 01011011 (0x5B)

N6

$0x\underbrace{DE}_{1}\underbrace{AD}_{2}\underbrace{BE}_{3}\underbrace{EF}_{4}$

So, Big-Endian convertion is $\underbrace{DE}_{1}\underbrace{AD}_{2}\underbrace{BE}_{3}\underbrace{EF}_{4}$ (since DE is the most significant byte

Little-Endian convertion is $\underbrace{EF}_{4}\underbrace{BE}_{3}\underbrace{AD}_{2}\underbrace{DE}_{1}$ (since EF is the least significant byte)

N7

5 bit:	extentended to 8 bit	sign-extended
7 = 00111	0000.0111	0000.0111
15 = 01111	0000.1111	0000.1111
-16 = 10000	000 10000	1111.0000
-5 = 11011	0001 1011	1111.1011

N8

7 = 0111

9 = 1001

4 = 0100

-5 = 1011

$$\begin{array}{r} 0111 \\ +\ 1001 \\ \hline 10000 \end{array} = 16$$

$$\begin{array}{r} 0100 \\ +\ 1011 \\ \hline 1111 \end{array} = -1$$